

Math 74: Algebraic Topology

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Problem 1.

- (a) Let $p : (E, e_0) \rightarrow (B, b_0)$ be a covering map. Suppose that E is path-connected. Then show that there is a surjection,

$$\phi : \pi_1(B, b_0) \rightarrow p^{-1}(b_0)$$

- (b) Moreover, if E is simply connected, then ϕ is a bijection.

Hint: Try to modify the computation of $\pi_1(S^1, (1,0))$ for the first and second part.

Solution.

- (a) For $[\gamma] \in \pi_1(B, b_0)$, let $\gamma' : I \rightarrow E$ be the unique lift of γ starting at e_0 . Define

$$\phi : \pi_1(B, b_0) \rightarrow p^{-1}(b_0), \quad \phi([\gamma]) = \gamma'(1).$$

Note $\gamma'(1) \in p^{-1}(b_0)$ since $p(\gamma'(1)) = \gamma(1) = b_0$, so ϕ is well-defined.

To see surjectivity, let $e_\alpha \in p^{-1}(b_0)$. Since E is path-connected, there exists a path $\sigma_\alpha : I \rightarrow E$ such that $\sigma_\alpha(0) = e_0$ and $\sigma_\alpha(1) = e_\alpha$.

Then $\gamma = p \circ \sigma_\alpha$ is a loop in B based at b_0 , since

$$\gamma(0) = (p \circ \sigma_\alpha)(0) = p(e_0) = b_0 \quad \text{and} \quad \gamma(1) = (p \circ \sigma_\alpha)(1) = p(e_\alpha) = b_0$$

By uniqueness of lifts, σ_α is the unique lift of γ starting at e_0 , so $\phi([\gamma]) = \sigma_\alpha(1) = e_\alpha$. We can similarly define a unique lift for each e_α and thus ϕ is surjective.

- (b) By part (a) we already know that ϕ is surjective. To see that ϕ is injective, let $[f], [g] \in \pi_1(B, b_0)$ such that $\phi([f]) = \phi([g]) = e_\alpha$ for some $e_\alpha \in p^{-1}(b_0)$, and let f', g' be the lifts of f, g starting at e_0 .

Then $f'(0) = g'(0) = e_0$ and $f'(1) = g'(1) = e_\alpha$. Since E is simply connected, $\pi_1(E, e_0) \cong \{e\}$, i.e. any two paths with the same initial and final points in E are

path-homotopic. Thus, $[f'] = [g']$, let F' be the homotopy between f' and g' . Then we have

$$F'(s, 0) = f'(s) \quad \text{and} \quad F'(s, 1) = g'(s).$$

Applying p to both sides,

$$(p \circ F')(s, 0) = p(f'(s)) = f(s) \implies p(f') = f$$

$$(p \circ F')(s, 1) = p(g'(s)) = g(s) \implies p(g') = g$$

Therefore $p \circ F' : I \times I \rightarrow B$ is a homotopy between f and g . Hence, $[f] = [g]$, ϕ is injective, and thus bijective.

Problem 2. Let $p : E \rightarrow B$ be a covering map, with B connected. Then if $p^{-1}(b_0)$ has k -many elements for some finite number k , $p^{-1}(b)$ also has k -many elements for every $b \in B$.

Solution.

Assume $p^{-1}(b_0)$ has k elements, and let U be an evenly covered neighborhood of b_0 , so that

$$p^{-1}(U) = \bigsqcup_{i=1}^k V_i$$

where each V_i maps homeomorphically onto U . For any $b \in U$, the preimage $p^{-1}(b)$ contains exactly one point from each V_i , hence has exactly k elements.

Now let $S = \{b \in B : |p^{-1}(b)| = k\}$ and $T = \{b \in B : |p^{-1}(b)| \neq k\}$, so $B = S \sqcup T$. We know that every point in S has an evenly covered neighborhood contained entirely in S , so S is open. Similarly, every point in T has an evenly covered neighborhood contained entirely in T , so T is open. Since B is connected and S is nonempty (as $b_0 \in S$), we must have $T = \emptyset$, i.e. $S = B$. Therefore $|p^{-1}(b)| = k$ for all $b \in B$.

Problem 3. If $p : E \rightarrow B$ is a covering map with E path-connected and B simply-connected. Then p is a homeomorphism.

Solution.

Let $p : E \rightarrow B$ be a covering map, where E is path-connected and B is simply-connected.

- **Surjectivity:** Covering maps are surjective by definition.
- **Injectivity:** Since B is simply-connected, $\pi_1(B, b_0) = \{e\}$ is trivial. By Problem 1(a), the map

$$\phi : \pi_1(B, b_0) \rightarrow p^{-1}(b_0), \quad [\gamma] \mapsto \tilde{\gamma}(1)$$

is surjective. Since the domain has exactly one element, $p^{-1}(b_0)$ has exactly one element. By Problem 2, this means that we have a 1-fold cover for B , i.e., $|p^{-1}(b)| = 1$ for all $b \in B$. Thus p is injective.

- **Continuity of p^{-1} :** Let $V \subseteq E$ be an open set. For each $e \in V$, let U_e be an evenly covered neighborhood of $p(e)$ in B . Since p is a 1-fold covering, $p^{-1}(U_e) = S_e$ is the unique disjoint set covering U_e , and

$$p|_{S_e} : S_e \rightarrow U_e$$

is a homeomorphism. Then $p(V \cap S_e) = p|_{S_e}(V \cap S_e)$ is open in $U_e \subseteq B$, and contains $p(e)$. Then,

$$p(V) = \bigcup_{e \in V} p(V \cap S_e)$$

is a union of open sets, hence open. Therefore p^{-1} is continuous.

Hence p is a continuous bijective map with a continuous inverse and thus a homeomorphism.

Problem 4. Show that if $p : E \rightarrow B$ and $p' : E' \rightarrow B'$ are covering maps, then so is $p \times p' : E \times E' \rightarrow B \times B'$, where $p \times p'$ is defined in an obvious manner.

Solution.

We define $p \times p' : E \times E' \rightarrow B \times B'$ by

$$(p \times p')(e, e') = (p(e), p'(e')).$$

This is continuous since each component is continuous.

Surjectivity. Given any $(b, b') \in B \times B'$, since p and p' are surjective there exist $e \in E$ and $e' \in E'$ with $p(e) = b$ and $p'(e') = b'$, so $(p \times p')(e, e') = (b, b')$.

Evenly covered. Let $(b, b') \in B \times B'$. Since p is a covering map, there exists an open neighborhood U of b in B that is evenly covered, with

$$p^{-1}(U) = \bigsqcup_{\alpha} V_{\alpha},$$

where each V_{α} is open in E and $p|_{V_{\alpha}} : V_{\alpha} \rightarrow U$ is a homeomorphism. Since p' is a covering map, there exists an open neighborhood U' of b' in B' that is evenly covered, with

$$p'^{-1}(U') = \bigsqcup_{\beta} V'_{\beta},$$

where each V'_{β} is open in E' and $p'|_{V'_{\beta}} : V'_{\beta} \rightarrow U'$ is a homeomorphism. Then $U \times U'$ is an open neighborhood of (b, b') in $B \times B'$, and

$$(p \times p')^{-1}(U \times U') = p^{-1}(U) \times p'^{-1}(U') = \left(\bigsqcup_{\alpha} V_{\alpha} \right) \times \left(\bigsqcup_{\beta} V'_{\beta} \right) = \bigsqcup_{\alpha, \beta} (V_{\alpha} \times V'_{\beta}).$$

Each $V_{\alpha} \times V'_{\beta}$ is open in $E \times E'$, and the restriction

$$(p \times p')|_{U_{\alpha} \times U'_{\beta}} = p|_{U_{\alpha}} \times p'|_{U'_{\beta}} : U_{\alpha} \times U'_{\beta} \rightarrow U \times U'$$

is a product of homeomorphisms, hence a homeomorphism. Hence $U \times U'$ is evenly covered, and $p \times p'$ is a covering map.

Problem 5. Show that if a subspace A is a deformation retraction of a space X , and another subspace B is a deformation retraction of A , then B is a deformation retraction of X .

Solution.

Let $H : X \times I \rightarrow X$ be the deformation retraction to A , so that $H(x, 0) = x$, $H(x, 1) \in A$, and $H(a, t) = a$ for all $x \in X$, $a \in A$, $t \in I$. Let $G : A \times I \rightarrow A$ be the deformation retraction to B , so that $G(a, 0) = a$, $G(a, 1) \in B$, and $G(b, t) = b$ for all $a \in A$, $b \in B$, $t \in I$.

Note that since $H(x, 1) \in A$ for all $x \in X$, the expression $G(H(x, 1), t)$ is well-defined for all $t \in I$. Now, we define the deformation retraction $P : X \times I \rightarrow X$ (from X to B) by,

$$P(x, t) = \begin{cases} H(x, 2t) & t \in [0, \frac{1}{2}] \\ G(H(x, 1), 2t - 1) & t \in [\frac{1}{2}, 1] \end{cases}$$

At $t = \frac{1}{2}$ we have $H(x, 1)$ and $G(H(x, 1), 0) = H(x, 1)$ respectively, so P is well-defined. Moreover, since both H and G are continuous, P is continuous by the pasting lemma.

We verify the three conditions for a deformation retraction to B :

- **Starts as identity:** $P(x, 0) = H(x, 0) = x$.
- **Ends in B :** $P(x, 1) = G(H(x, 1), 1) \in B$.
- **B is fixed throughout:** Let $b \in B \subseteq A$. For $t \in [0, \frac{1}{2}]$, $P(b, t) = H(b, 2t) = b$ since $b \in A$. For $t \in [\frac{1}{2}, 1]$, $P(b, t) = G(H(b, 1), 2t - 1) = G(b, 2t - 1) = b$ since $b \in B$.

Hence, P is a deformation retraction from X to B .

Problem 6.

- (a) Show that if a space retracts onto a subspace A , then the homomorphism on the fundamental group induced by the inclusion $\iota : A \hookrightarrow X$ is injective.
- (b) Also show that if A is a deformation retract of X , then ι_* is an isomorphism.

Solution.

- (a) Let r be the map that retracts X to A , and $r(x_0) = a_0$ in A . Note that the composition $r \circ \iota$ is the identity on A . Now consider the homomorphism $(r \circ \iota)_*$. We have,

$$(r \circ \iota)_* = r_* \circ \iota_* = (\text{id}_A)_* = \text{id}_{\pi_1(A, a_0)}$$

Let $\iota_*([f]) = \iota_*([g])$. Then,

$$\begin{aligned} r_*(\iota_*([f])) &= r_*(\iota_*([g])) \\ \implies (\text{id}_A)_*([f]) &= (\text{id}_A)_*([g]) \\ \implies [f] &= [g] \end{aligned}$$

Hence ι_* is injective.

- (b) Let H be the deformation retraction of X to A , and $H(x_0, 1) = a_0$ in A . From part (a) we know that ι_* is injective. So we only need to show that ι_* is surjective.

We define the continuous map $r = H(x, 1)$. Consider the map $\iota \circ r$ which maps $X \rightarrow A \rightarrow X$. This map is homotopic to the identity map by H (by definition),

$$\iota \circ r \simeq \text{id}_X$$

Then $(\iota \circ r)_* = \iota_* \circ r_* \simeq (\text{id}_X)_* = \text{id}_{\pi_1(X, x_0)}$ (Lemma 58.1). Since $\text{im}(r_*) = \pi_1(A, a_0)$, ι_* is surjective. Thus, we have an isomorphism $\pi_1(A, a_0) \cong \pi_1(X, x_0)$.

Problem 7. Show that if a space X deformation retracts to a point $x \in X$, then for each neighborhood U of x in X there exists a neighborhood $V \subset U$ of x such that the inclusion map $V \hookrightarrow U$ is nullhomotopic (nullhomotopic means homotopic to a constant map).

Solution.

Let $H : X \times I \rightarrow X$ be the deformation retraction from X to $\{x\}$ such that $\forall x' \in X$,

$$H(x', 0) = x', \quad H(x', 1) = x, \quad \text{and} \quad H(x, t) = x$$

and let $U \subseteq \text{im}(H) = X$ be an open set containing x . Note that the preimage $H^{-1}(U)$ is open in $X \times I$ and contains $\{x\} \times I$ (since $x \in U$).

We claim there exists an open neighborhood V of x in X such that $V \times I \subseteq H^{-1}(U)$.

For each $t \in I$, continuity of H at (x, t) gives a open set $V_t \times W_t$ containing (x, t) with $H(V_t \times W_t) \subseteq U$. The sets $\{W_t\}_{t \in I}$ form an open cover of I . Since I is compact, there is a finite subcover $W_{t_1}, \dots, W_{t_n}$. Let $V = V_{t_1} \cap \dots \cap V_{t_n}$ (finite intersection of open sets containing x is an open set containing x). Note that for any $(y, t) \in V \times I$, we have $t \in W_{t_k}$ for some k and $y \in V \subseteq V_{t_k}$, so $H(y, t) \in U$.

Hence there exists some open set $V \times I \subseteq H^{-1}(U)$ and the restriction $H|_{V \times I} : V \times I \rightarrow U$ is a homotopy from the inclusion $\iota : V \hookrightarrow U$ (at $t = 0$) to the constant map at x (at $t = 1$) so $\iota \simeq e_x$.

Problem 8.

- (a) Let X be the subspace of $\mathbb{R}^2$ consisting of the horizontal segment $[0, 1] \times 0$ together with all the vertical segments $r \times [0, 1 - r]$ for r a rational number in $[0, 1]$. Show that X deformation retracts to any point in the segment $[0, 1] \times 0$, but not to any other point. [See the preceding problem.]
- (b) Let Y be the subspace of $\mathbb{R}^2$ that is the union of an infinite number of copies of X arranged as in the figure below. Show that Y is contractible but does not deformation retract onto any point.



Solution.

- (a) Let $\mathbf{a} = (x_0, 0)$ be an arbitrary point in $[0, 1] \times \{0\}$. It suffices to show that X deformation retracts to the subspace $B = [0, 1] \times \{0\}$, since B is convex and hence deformation retracts to any of its points via the straight-line homotopy, and by Problem 5 the composition of deformation retracts is a deformation retract. We define $H : X \times I \rightarrow X$ as follows:

$$H((x, y), t) = (1 - t)(x, y) + t(x, 0)$$

Note that

$$H((x, y), 0) = (x, y) \in X, \quad H((x, y), 1) = (x, 0) \in B$$

And

$$H((x, 0), t) = (1 - t)(x, 0) + t(x, 0) = (x, 0) \in B$$

So H is a deformation retraction from X to B . Since $B \cong I$ is convex, it deformation retracts to $\mathbf{a}$ via

$$G((x, 0), t) = ((1 - t)x + tx_0, 0).$$

By Problem 5, X deformation retracts to $\{\mathbf{a}\}$.

We show that X does not deformation retract to any point $\mathbf{a} = (r, h)$, $h > 0$ by showing the contrapositive of the condition in Problem 7. That is there exists some neighborhood $U \subseteq X$ of $\mathbf{x}$ such that for all neighborhoods $V \subset U$ of $\mathbf{x}$, the inclusion map is not nullhomotopic then X does not deformation retract to $\mathbf{a} \in X$.

Define $0 < \varepsilon < \min(h, 1 - r)$ and let U be the ε -ball centered at $\mathbf{a}$ in X ,

$$U = B_\varepsilon(\mathbf{a}) \cap X$$

That is, $\mathcal{U}$ does not intersect the x -axis and hence is composed of infinitely many disjoint line segments (one for each rational x with $|x - r| < \varepsilon$.)

$$\mathcal{U} = \bigsqcup_{\substack{r \in \mathbb{Q} \\ |x-r| < \varepsilon}} (\{r\} \times (h - \varepsilon, h + \varepsilon)) \cap X.$$

Now let $V \subseteq \mathcal{U}$ be any sub-neighborhood of $\mathbf{a}$. Then V is also composed of infinitely many disjoint line segments.

We claim the inclusion $\iota : V \hookrightarrow \mathcal{U}$ is not nullhomotopic. If it were, there would exist a homotopy

$$H : V \times I \rightarrow \mathcal{U}$$

with $H(v, 0) = v$ and $H(v, 1) = \mathbf{u}_0$ for some fixed $\mathbf{u}_0 \in \mathcal{U}$.

In particular, for each $v \in V$, there must be some path connecting v to $\mathbf{u}_0$ inside $\mathcal{U}$. But v and $\mathbf{u}_0$ may lie in different path components of $\mathcal{U}$, making this impossible. Hence there exists a $\mathcal{U}$ such that for all $V \subset \mathcal{U}$ the inclusion map is not nullhomotopic and thus X does not deformation retract to $\mathbf{a} \in X$.

(b) **TODO: contractible**

We show that Y does not deformation retract to any point. By part (a), within each copy of X , the only candidates for a deformation retract are points on the base x -axis. The only points that qualify are those that are on the corner of two adjacent copies of X , since they lie on the base of both X s. So it suffices to show that Y does not deformation retract to any such $\mathbf{a}$.

For the sake of contradiction assume it retracts to a point $\mathbf{a} = (x_0, y_0)$. WLOG let $\mathbf{a} = (0, 1)$. Define $\varepsilon > 0$ and let $\mathcal{U}$ be the ε -ball centered at $\mathbf{a}$ in Y ,

$$\mathcal{U} = B_\varepsilon(\mathbf{a}) \cap Y$$

That is, $\mathcal{U}$ is composed of infinitely many disjoint line segments. Now let $V \subseteq \mathcal{U}$ be any sub-neighborhood of $\mathbf{a}$. Then V contains an open ball $B_\delta(\mathbf{a})$ for some $\delta > 0$. Consider the sequence of points

$$\mathbf{a}_n = \left(\frac{1}{n}, 1 - \frac{1}{n} \right).$$

Each $\mathbf{a}_n$ lies in Y since $1/n$ is rational, and each $\mathbf{a}_n$ is the tip of a distinct vertical segment, hence lies in a distinct path component of $\mathcal{U}$. Since $\lim_{n \rightarrow \infty} \mathbf{a}_n \rightarrow \mathbf{a}$, the tail of the sequence eventually lies in $B_\delta(\mathbf{a}) \subseteq V$.

Suppose $\iota : V \hookrightarrow \mathcal{U}$ were nullhomotopic. Then there would be a homotopy $H : V \times I \rightarrow \mathcal{U}$ with $H(v, 0) = v$ and $H(v, 1) = \mathbf{u}_0$ for some fixed $\mathbf{u}_0 \in \mathcal{U}$ and all $v \in V$. In particular, for each $v \in V$, there must be some path connects v to $\mathbf{u}_0$ inside $\mathcal{U}$. But for large enough n , $\mathbf{a}_n \in V$ lies in a different path component of $\mathcal{U}$ than $\mathbf{u}_0$, so no such path can exist. This is a contradiction, so no such V exists satisfying the conclusion of Problem 7. Therefore Y does not deformation retract to $\mathbf{a}$.

Problem 9. Define $f : S^1 \times I \rightarrow S^1 \times I$ by $f(\theta, s) = (\theta + 2\pi s, s)$, so f restricts to the identity on the two boundary circles of $S^1 \times I$. Show that f is homotopic to the identity by a homotopy f_t that is stationary on one of the boundary circles, but not by any homotopy f_t that is stationary on both boundary circles. [Consider what f does to the path $s \mapsto (\theta_0, s)$ for fixed $\theta_0 \in S^1$].

Solution.

To show that f is homotopic to the identity, we define $f_t : S^1 \times I \rightarrow S^1 \times I$ as follows:

$$f_t(\theta, s) = (\theta + 2\pi s(1 - t), s)$$

We check,

$$f_0(\theta, s) = (\theta + 2\pi s, s) = f, \quad f_1(\theta, s) = (\theta, s) = \text{id}, \quad f_t(\theta, 0) = (\theta, 0) = \text{id}|_{S^1 \times \{0\}}$$

Hence, f is homotopic to the identity by a homotopy f_t that is stationary on one of the boundary circles.

Now suppose for contradiction that there exists a homotopy $f_t : S^1 \times I \rightarrow S^1 \times I$ between f and the identity that is stationary on both boundary circles. Let θ_0 be some point in S^1 and consider the vertical path

$$\gamma : I \rightarrow S^1 \times I, \quad \gamma(s) = (\theta_0, s).$$

Since f_t is stationary on both boundary circles, we have $f_t(\theta_0, 0) = (\theta_0, 0)$ and $f_t(\theta_0, 1) = (\theta_0, 1)$ for all t . Therefore $f_t \circ \gamma$ is a homotopy between γ and $f \circ \gamma$, where

$$(f \circ \gamma)(s) = (\theta_0 + 2\pi s, s).$$

Now let $\pi : S^1 \times I \rightarrow S^1$ be the projection $\pi(\theta, s) = \theta$. Applying π , we get a homotopy in S^1 between

$$\pi \circ \gamma \equiv \theta_0 \quad (\text{the constant path})$$

and

$$\pi \circ f \circ \gamma : s \mapsto \theta_0 + 2\pi s \quad (\text{a loop winding once around } S^1).$$

But these two loops represent different elements of $\pi_1(S^1, \theta_0) \cong \mathbb{Z}$ (0 and 1 respectively). Since homotopic loops represent the same element of π_1 this is a contradiction! Hence, no such homotopy f_t exists.